

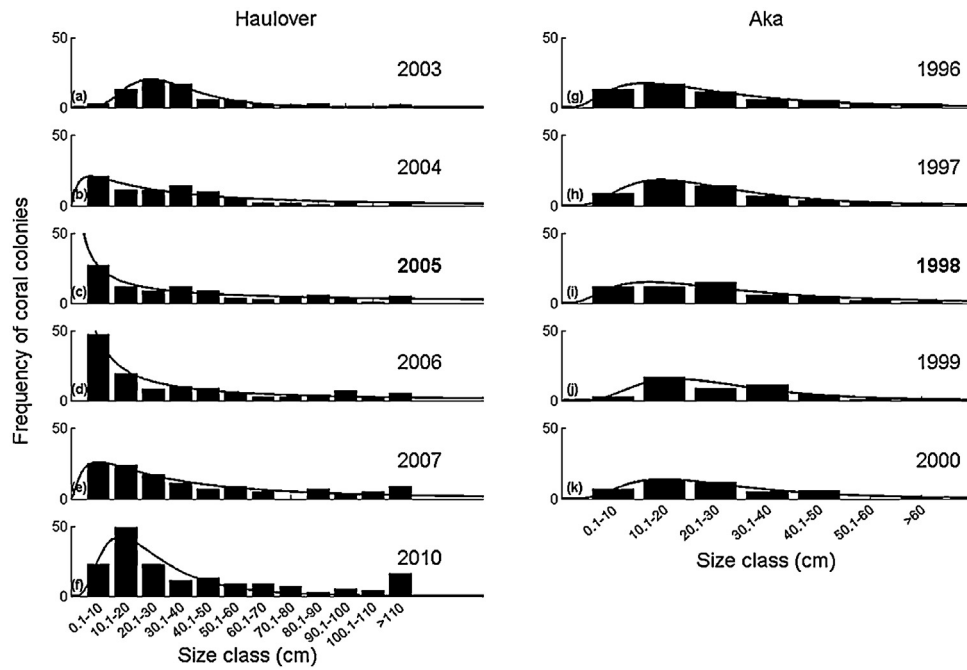


Fig. 5. (a–k) The size-frequency of field data for Haulover Bay, US Virgin Islands, using 12 size classes (a–f), and Aka Jima, Japan, using 7 size classes (g–k). Each size class ranges 10 cm. The abundances are for the entire 500 m × 40 m area in Haulover Bay and the quadrat area of 2 m × 2 m in Aka Jima. Bold font indicates the 2005 thermal anomaly in Haulover Bay (c) and the 1998 thermal anomaly in Aka Jima (i).

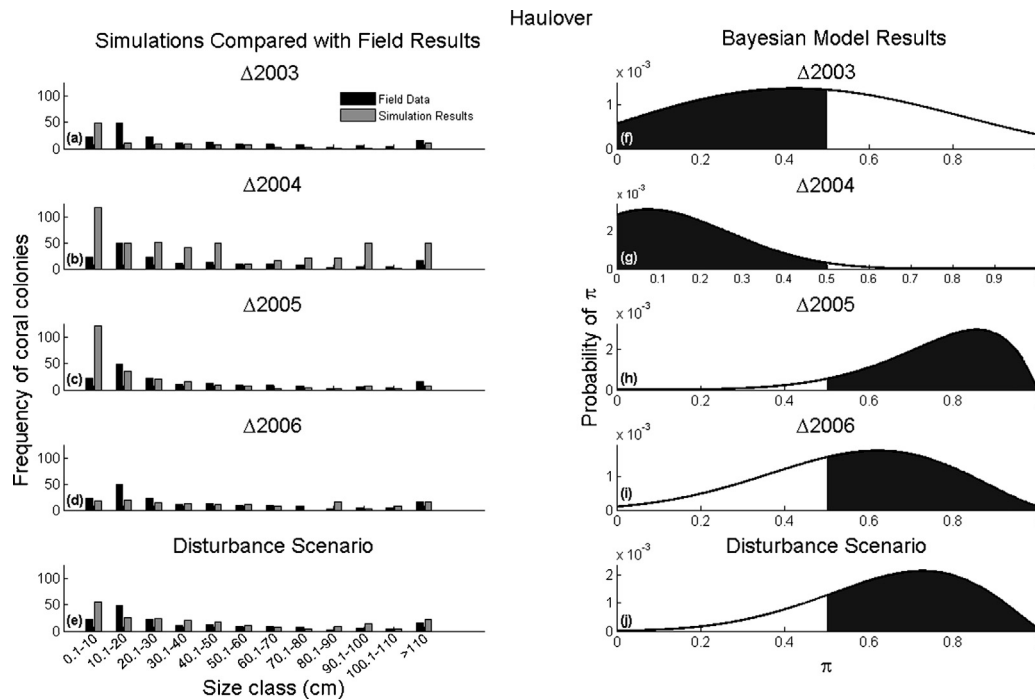


Fig. 6. (a–j) Haulover Bay, US Virgin Islands, size-frequency distributions of field data compared with simulations. The indicated years represent the field data used for each simulation in (a–d). The disturbance scenario is depicted in (e). The disturbance scenario uses an average matrix of all the data matrices (excluding the thermal anomaly year of 2005) for every year except for years 2–3. The data for years 2–3 comes from the matrices for the thermal anomaly in 2005 at Haulover Bay (Table 1). All simulations were run until year 2010 to determine size class abundances for comparison with field data for that year. The Bayesian outputs (f–j) indicate which model type best describes the two distributions (where $\pi > 0.5$ is “over-lapping distribution model” and $\pi < 0.5$ is “distinct distributions model”). The maximum π values for each simulation are (f) 0.423, (g) 0.075, (h) 0.856, (i) 0.621, (j) 0.728. Transitional data are indicated by the delta symbol (Δ) which represents a change from the listed year to the following year (i.e. $\Delta 2003$ represents the transition from 2003 to 2004).